

CHAPTER 22: ELECTRIC FIELDS – EXERCISES

Multiple choice questions:

1. A piece of plastic has a *net* charge of $+2.00\ \mu\text{C}$. How many more protons than electrons does this piece of plastic have?

A. 3.20×10^{13}
B. 3.20×10^{19}
C. 1.25×10^{19}
D. 1.25×10^{13}
E. 8.00×10^{12}

Ans: D

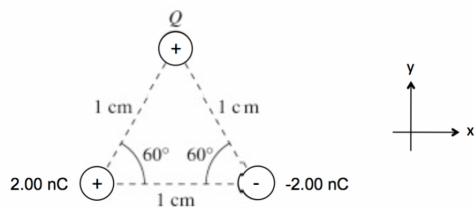
2. Three point charges are placed on the x -axis. The first charge of $+2.0\ \mu\text{C}$ is located at the origin, the second charge of $-2.0\ \mu\text{C}$ is located at $x = 50\ \text{cm}$, and the third charge of $+4.0\ \mu\text{C}$ is located at $x = 100\ \text{cm}$. Find the electrostatic force that acts on the charge located at the origin.

A. $-0.036\hat{\mathbf{i}}\ \text{N}$
B. zero
C. $0.072\hat{\mathbf{i}}\ \text{N}$
D. $0.036\hat{\mathbf{i}}\ \text{N}$
E. $-0.072\hat{\mathbf{i}}\ \text{N}$

Ans: C

3. Consider the three point charges arranged as shown in the figure below. Find the electrostatic force acting on $Q = 6.00\ \text{nC}$.

Hint: $1\ \text{nC} = 10^{-9}\ \text{C}$



A. $-2.16 \times 10^{-3}\hat{\mathbf{j}}\ \text{N}$
B. $2.16 \times 10^{-3}\hat{\mathbf{j}}\ \text{N}$
C. $-1.08 \times 10^{-3}\hat{\mathbf{i}}\ \text{N}$
D. $1.08 \times 10^{-3}\hat{\mathbf{j}}\ \text{N}$
E. $1.08 \times 10^{-3}\hat{\mathbf{i}}\ \text{N}$

Ans: E

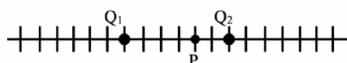
4. Particle 1 with charge q_1 and particle 2 with charge q_2 are on the x -axis, with particle 1 located at $x = a$ and particle 2 located at $x = -2a$. For the *net* electrostatic force on a third charged particle located at the origin to be zero, q_1 and q_2 must be related by

A. $q_2 = -2q_1$
B. $q_2 = -4q_1$
C. $q_2 = 2q_1$
D. $q_2 = q_1 / 4$
E. $q_2 = 4q_1$

Ans: E

5. Two point charges Q_1 and Q_2 are located as shown in the figure below. The electric field is observed to be *zero* at point P. Which of the following statements are *correct*?

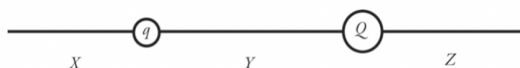
- I. The magnitudes of Q_1 and Q_2 are equal
- II. The magnitude of Q_1 is greater than the magnitude of Q_2
- III. The magnitude of Q_1 is less than the magnitude of Q_2
- IV. Q_1 and Q_2 have the same sign
- V. Q_1 and Q_2 have opposite signs



- A. II and V
- B. III and IV
- C. III and V
- D. II and IV
- E. I and IV

Ans: D

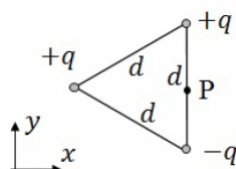
6. Two *unequal* point charges, q and Q , are located along the straight line shown in the figure below. The charges have *opposite* signs and $|Q| > |q|$. On which of the linear segments labeled X, Y, and Z can there be a point at which the *net* electric field due to these two charges will be zero?



- A. only Z
- B. only X
- C. all three segments
- D. only Y
- E. only X and Z

Ans: B

7. As shown in the figure below, three point charges are located at the vertices of an equilateral triangle with side length d . Find the *net* electric field created by all these three charges at point P.



- A. $\frac{4kq}{3d^2} \hat{\mathbf{i}}$
- B. $\frac{4kq}{d^2} \left(\frac{1}{3} \hat{\mathbf{i}} + 2\hat{\mathbf{j}} \right)$
- C. $\frac{kq}{d^2} \left(\frac{1}{3} \hat{\mathbf{i}} + 2\hat{\mathbf{j}} \right)$
- D. $\frac{4kq}{d^2} \left(\frac{1}{3} \hat{\mathbf{i}} - 2\hat{\mathbf{j}} \right)$
- E. $\frac{kq}{d^2} \left(\frac{1}{3} \hat{\mathbf{i}} - 2\hat{\mathbf{j}} \right)$

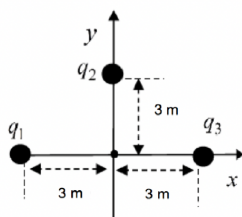
Ans: D

8. A very small ball has a mass of 8 g and a charge of $-4 \mu\text{C}$. Find the magnitude and direction of the electric field that will balance the weight of the ball so that the ball is suspended motionless in air.

- A. $8 \times 10^3 \text{ N/C}$, vertically upward
- B. $2 \times 10^4 \text{ N/C}$, vertically upward
- C. $4 \times 10^5 \text{ N/C}$, vertically upward
- D. $2 \times 10^4 \text{ N/C}$, vertically downward
- E. $8 \times 10^3 \text{ N/C}$, vertically downward

Ans: D

9. Three point charges $q_1 = q_2 = +2 \mu\text{C}$ and $q_3 = -2 \mu\text{C}$ are located in the xy -plane as shown in the figure below. Find the *net* electric field at the origin.



- A. $(-4\hat{i} - 2\hat{j}) \text{ kN/C}$
- B. $(12\hat{i} + 6\hat{j}) \text{ kN/C}$
- C. $(12\hat{i} - 6\hat{j}) \text{ kN/C}$
- D. $(4\hat{i} + 2\hat{j}) \text{ kN/C}$
- E. $(4\hat{i} - 2\hat{j}) \text{ kN/C}$

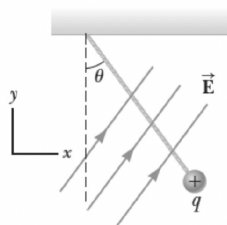
Ans: E

10. Suppose that there exists a uniform electric field of 4 N/C directed along the *positive* x -axis. When a point charge is placed at and fixed to the origin, the resulting electric field on the x -axis at $x = 2 \text{ m}$ becomes zero. What is then the magnitude of the resulting electric field on the x -axis at $x = 4 \text{ m}$?

- A. 1 N/C
- B. 3 N/C
- C. 4 N/C
- D. More information is needed to find the magnitude of the resulting electric field at this position
- E. 2 N/C

Ans: B

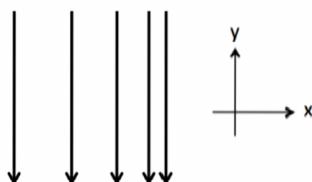
11. As shown in the figure below, a charged ball of mass 2 g is suspended on a light string in the presence of a uniform electric field of $\vec{E} = (3 \times 10^5 \hat{i} + 6 \times 10^5 \hat{j}) \text{ N/C}$. The ball is in equilibrium at $\theta = 37^\circ$. Determine the charge on the ball.



- A. $5 \times 10^{-8} \text{ C}$
- B. $2 \times 10^{-8} \text{ C}$
- C. $5 \times 10^{-6} \text{ C}$
- D. $3 \times 10^{-8} \text{ C}$
- E. $4 \times 10^{-6} \text{ C}$

Ans: B

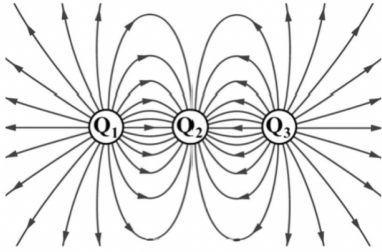
12. The figure below shows the electric field lines in a given region of space. The magnitude of the field



- A. decreases as x increases
- B. decreases as y increases
- C. is constant
- D. increases as x increases
- E. increases as y increases

Ans: D

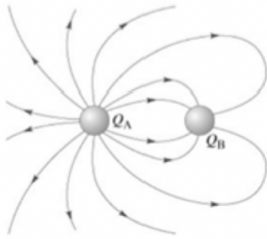
13. The figure below shows three electric charges labeled Q_1 , Q_2 , and Q_3 , and some electric field lines in the region surrounding the charges. What are the signs of these three charges?



- A. All three charges are positive
- B. Q_1 is positive, Q_2 is positive, and Q_3 is negative
- C. Q_1 is negative, Q_2 is positive, and Q_3 is negative
- D. All three charges are negative
- E. Q_1 is positive, Q_2 is negative, and Q_3 is positive

Ans: E

14. The figure below shows the electric field lines for a system of two point charges Q_A and Q_B . Which of the following could represent the magnitudes and signs of Q_A and Q_B ? Take q to be a positive quantity.



- A. $Q_A = -3q$, $Q_B = +7q$
- B. $Q_A = +3q$, $Q_B = -7q$
- C. $Q_A = +7q$, $Q_B = -3q$
- D. $Q_A = -7q$, $Q_B = +3q$
- E. $Q_A = +q$, $Q_B = -q$

Ans: C

15. A particle of mass 4×10^{-15} kg and charge -5×10^{-8} C is projected into a region where the electric field is $8 \times 10^3 \hat{j}$ N/C. If the initial velocity of the particle is $(4 \times 10^5 \hat{i} + 2 \times 10^5 \hat{j})$ m/s, what will be its speed after moving for 5×10^{-6} s in this region?

- A. 10×10^5 m/s
- B. 3×10^5 m/s
- C. 8×10^5 m/s
- D. 4×10^5 m/s
- E. 5×10^5 m/s

Ans: E

Additional problems:

Two small metallic spheres, each of mass $m = (0.200 \pm 0.001) \text{ g}$, are suspended as pendulums by light strings of length L as shown in **Figure P23.6**. The spheres are given the same electric charge of $(7.2 \pm 0.5) \text{ nC}$, and they come to equilibrium when each string is at an angle of $\theta = (5.00 \pm 0.05)^\circ$ with the vertical. How long are the strings?

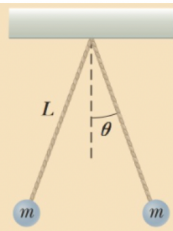


Figure P23.6

0.299 m

Four charged particles are at the corners of a square of side a as shown in **Figure P23.17**. Determine (a) the electric field at the location of charge q and (b) the total electrostatic force exerted on q .

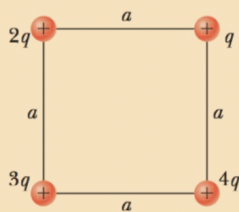


Figure P23.17

- (a) $5.91 \frac{kq}{a^2}$ (58.8° counterclockwise above a horizontal line),
 (b) $5.91 \frac{kq^2}{a^2}$ (58.8° counterclockwise above a horizontal line)

A proton is projected in the positive x direction into a region of a uniform electric field $\vec{E} = (-6.00 \times 10^5) \hat{i} \text{ N/C}$ at $t = 0$. The proton travels 7.00 cm as it comes to rest. Determine (a) the acceleration of the proton, (b) its initial speed and (c) the time interval over which the proton comes to rest.

- (a) $-5.75 \times 10^{13} \hat{i} \text{ m/s}^2$, (b) $2.84 \times 10^6 \text{ m/s}$, (c) $4.94 \times 10^{-8} \text{ s}$